

ASSIGNMENT 9.5

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Batch 24

AIAC

Problem 1: String Utilities Function

Consider the following Python function:

```
def reverse_string(text):
```

```
    return text[::-1]
```

Task:

1. Write documentation in:

- o (a) Docstring
- o (b) Inline comments
- o (c) Google-style documentation

2. Compare the three documentation styles.

3. Recommend the most suitable style for a utility-based string library.

```
C:\> Users > ADMIN > OneDrive > Desktop > AIAC > string_utils.py > ...  
1  from random import sample  
2  def reverse_string_doc(text):  
3      """  
4      Reverse the given string.  
5      Parameters:  
6      |   text (str): The input string.  
7      Returns:  
8      |   str: Reversed string.  
9      """  
10     return text[::-1]  
11  
12     def reverse_string_inline(text):  
13         # This function reverses a string  
14         # using slicing method  
15         return text[::-1]  
16  
17     def reverse_string_google(text):  
18         """  
19         Reverses the given string.  
20         Args:  
21         |   text (str): Input string.  
22         Returns:  
23         |   str: Reversed string.  
24         """  
25         return text[::-1]
```

OUTPUT:

```
PS C:\Users\ADMIN> & "C:/Program Files/Python313/python.exe" c:/Users/ADMIN/OneDrive/Desktop/AIAC/g_utils.py
olleH
olleH
olleH
PS C:\Users\ADMIN>
```

Problem 2: Password Strength Checker

Consider the function:

```
def check_strength(password):
    return len(password) >= 8
```

Task:

1. Document the function using docstring, inline comments, and Google style.
2. Compare documentation styles for security-related code.
3. Recommend the most appropriate style.

```
C:\Users\ADMIN\OneDrive\Desktop\AIAC> passwordstrengthchecker.py
1 def password_strength_checker(password):
2     """Checks the strength of a given password.
3     A password is considered strong if it length is at least 8 characters
4     parameters:
5     | password (str): The password to check.
6     returns:
7     | str: A message indicating the strength of the password.
8     """
9     return len(password) >= 8
10
11 def password_strength_cheker(password):
12     # This function checks if the password is strong
13     # A password is considered strong if it has at least 8 characters
14     return len(password) >= 8
15
16 def password_strength_checker(password):
17     """
18     check wheather the password meets the strength requirements.
19     Args:
20     | password (str): The password to check.
21     Returns:
22     | bool: True if the password is strong, False otherwise.
23     """
24     return len(password) >= 8
25
```

OUTPUT:

```
> v TERMINAL
PS C:\Users\ADMIN> & "C:/Program Files/Python313/python.exe" c:/Users/ADMIN/OneDrive/Desktop/AIAC/passwordstrengthchecker.py
True
True
True
```

Problem 3: Math Utilities Module

Task:

1. Create a module math_utils.py with functions:
 - o square(n)
 - o cube(n)
 - o factorial(n)
2. Generate docstrings automatically using AI tools.
3. Export documentation as an HTML file.

```
C: > Users > ADMIN > OneDrive > Desktop > AIAC > math_utils.py > ...  
1  def square(x):  
2      """Returns the square of a number.  
3      Parameters:  
4      x (int or float): The number to be squared.  
5      Returns:  
6      int or float: The square of the input number.  
7      """  
8      return x * x  
9  def cube(x):  
10     """Returns the cube of a number.  
11     Parameters:  
12     x (int or float): The number to be cubed.  
13     Returns:  
14     int or float: The cube of the input number.  
15     """  
16     return x * x * x  
17  def factorial(n):  
18     """Returns the factorial of a number.  
19     Parameters:  
20     n (int): The number to calculate the factorial of. Must be a non-negative integer.  
21     Returns:  
22     int: The factorial of the input number.  
23     """  
24     if n == 0 or n == 1:
```

OUTPUT:

```
PS C:\Users\ADMIN> & "C:/Program Files/Python313/python.exe" c:/Users/ADMIN/OneDrive  
utils.py  
int or float: The cube of the input number.  
  
Returns the factorial of a number.  
Parameters:  
n (int): The number to calculate the factorial of. Must be a non-negative integer.  
Returns:  
int: The factorial of the input number.
```

Problem 4: Attendance Management Module

Task:

1. Create a module attendance.py with functions:
 - o mark_present(student)
 - o mark_absent(student)

- o get_attendance(student)
2. Add proper docstrings.
3. Generate and view documentation in terminal and browse

```
C:\Users\ADMIN> OneDrive\Desktop\AIAC> attendance.py mark_present

1 #attendance.py
2 """
3 Attendance management module
4 this module provides functions to manage attendance records for students in a school or university setting.
5 """
6 #dictionary to store attendance records
7 attendance_records = {}
8 def mark_present(student):
9     """
10     Marks a student as present in the attendance records.
11     Args:
12         student (_type_): _description_
13     Returns:
14         _type_: _description_
15     """
16     attendance_records[student] = 'Present'
17 def mark_absent(student):
18     """Marks a student as absent in the attendance records.
19     Args:
20         student (_type_): _description_
21     Returns:
22         _type_: _description_
23     """
24     attendance_records[student] = 'Absent'
25 def get_attendance(student):
```

OUTPUT:

```
PS C:\Users\ADMIN\OneDrive\Desktop\AIAC> python -m pydoc attendance
Help on module attendance:

NAME
    attendance

DESCRIPTION
    Attendance management module
    this module provides functions to manage attendance records for students in a school or university setting.

PS C:\Users\ADMIN\OneDrive\Desktop\AIAC> python -m pydoc -b
Server ready at http://localhost:56348/
Server commands: [b]rowser, [q]uit
```

attendance [c:\users\admin\onedrive\desktop\aiac\attendance.py](http://localhost:56348/attendance.py)

Attendance management module
this module provides functions to manage attendance records for students in a school or university setting.

Functions

```
get_attendance(student)
    Retrieves the attendance status of a student.
    Args:
        student (_type_): _description_
    Returns:
        _type_: _description_

mark_absent(student)
    Marks a student as absent in the attendance records.
    Args:
        student (_type_): _description_
    Returns:
        _type_: _description_

mark_present(student)
    Marks a student as present in the attendance records.
    Args:
        student (_type_): _description_
    Returns:
        _type_: _description_

Data
    attendance_records = {}
```

Problem 5: File Handling Function

Consider the function:

```
def read_file(filename):  
    with open(filename, 'r') as f:  
        return f.read()
```

Task:

1. Write documentation using all three formats.
2. Identify which style best explains exception handling.
3. Justify your recommendation.

```
C:\> Users > ADMIN > OneDrive > Desktop > AIAC > file_utils.py > ...  
1  def read_file_doc(filename):  
2      """  
3      read the contents of the file and return it as a string.  
4      Parameters:  
5          filename (str): The name of the file to read.  
6      Returns:  
7          str: The contents of the file.  
8      raises:  
9          FileNotFoundError: If the file does not exist.  
10         IOError: If there is an error reading the file.  
11      """  
12     with open(filename, 'r') as file:  
13         return file.read()  
14  
15     def read_file_inline(filename):  
16         #open the file in read mode and return its contents as a string  
17         #if the file does not exist, a FileNotFoundError will be raised  
18         with open(filename, 'r') as file:  
19             #read the contents of the file and return it  
20             return file.read()  
21     def read_file_google(filename):  
22         """  
23         Reads the contents of a file and returns it as a string.  
24         Args:  
25             filename (str): The name of the file to read.
```

OUTPUT:

```
PS C:\Users\ADMIN\OneDrive\Desktop\AIAC> & "C:/Program Files/Python313/python.exe" c:/Users/ADMIN/OneDrive/Desktop/AIAC/file_utils.py  
  
read the contents of the file and return it as a string.  
Parameters:  
    filename (str): The name of the file to read.  
Returns:  
    str: The contents of the file.    raises:  
    FileNotFoundError: If the file does not exist.  
    IOError: If there is an error reading the file.  
  
None  
  
Reads the contents of a file and returns it as a string.  
Args:  
    filename (str): The name of the file to read.  
Returns:  
    str: The contents of the file.  
Raises:  
    FileNotFoundError: If the file does not exist.  
    IOError: If there is an error reading the file.
```